DLD Homework 2 Due date : October 17, 2025 – 23:00

1. Given the Boolean algebra equation: F(A,B,C) = BC’ + AB’ + A’BC
   a) Give the truth table for F

	A	B	C	F
0	0	0	0	0
1	0	0	1	0
2	0	1	0	1
3	0	1	1	1
4	1	0	0	1
5	1	0	1	1
6	1	1	0	1
7	1	1	1	0

01	01	100	0	0	0
00	01	101	0	0	0
11	01	110	1	0	0
10	01	111	0	0	1
01	11	000	0	1	0
00	11	001	0	1	0
11	10	010	1	0	0
10	10	011	0	0	0

   b) Give the canonical sum of minterms ($\sum(\boldsymbol{i})$) representation for F
   c) Give the canonical sum of products (SOP) Boolean representation for F
   d) Give the canonical product of maxterms ($\Pi\ (\boldsymbol{i})$) representation for F
   e) Give the canonical product of sums (POS) Boolean representation for F

2. The duality principle states that any theorem or identity in Boolean algebra is also true if 0s and 1s and then AND and OR operators are swapped throughout.
   a) True b) False

3. Perform the following additions in **8421 BCD**. Show steps and correct invalid results (> 1001_2) by adding 0110_2. Express your final result in both **BCD** and **decimal**.
   (a) $0101_{BCD} + 0011_{BCD}$
   (b) $1001_{BCD} + 0100_{BCD}$
   (c) $0101\_0111_{BCD} + 0010\_1001_{BCD}$
   (d) $0111\_1001\_1000_{BCD} + 0001\_0100\_0100_{BCD}$

4. A 3-bit absolute shaft encoder is designed to represent 8 angular positions of a shaft (0°–315° in steps of 45°). (a) Construct a table showing the shaft positions, their **8421 binary representation**, and the corresponding **Gray code**. (b) Explain why Gray code is preferred over binary code for this application. (c) For the 8421 Binary coding, when going from 011 → 100, list all intermediate invalid states, i.e., the “wrong” values that could appear (depending on which sensor changes first).

Shaft Angle	Binary (8421)	Gray Code
0	000	000
45	001	001
90	010	011
135	011	010
180	100	110
225	101	111
270	110	101
315	111	100

✓

6 4 2 4 0 0 7 2

5. Assume your student ID is an 8-digit decimal number $\boldsymbol{d_7d_6d_5d_4d_3d_2d_1d_0}$
   (a) Convert the real number $\boldsymbol{d_6\,d_0\,.\,d_3\,d_7}$ to octal. For the octal number use at least 2 fractional digits.
   (b) Convert the octal number from part (a) to binary.
   (c) Convert the binary number from part (b) to hexadecimal.

Question 3)

a)
$$\begin{array}{r} {\scriptstyle +1\ +1\ +1} \\ 0101 \\ +\ 0011 \\ \hline 1000 \end{array} \to 1000_{BCD} \to 8_{10}$$

b) $1001_{BCD} + 0100_{BCD} \to 9+4 = 13_{10}$

$$\begin{array}{r} 1001 \\ +\ 0100 \\ \hline 1101 \end{array} \to 1101_2\ (13)$$

$$\begin{array}{r} 1101 \\ +\ 0110 \\ \hline 1\ 0011 \end{array} \to 0001\ \ 0011_{BCD} \to 13_{10}$$

c) $0101\ 0111_{BCD} + 0010\ 1001_{BCD}$

$= 57 + 29 = 86_{10}$

$$
\begin{array}{rr}
 & 0101\ 0111 \\
+ & 0010\ 1001 \\
\hline
 & 1000\ 0000 \\
 & 0110 \\
\hline
+ & 1000\ 0110_{BCD}
\end{array}
\rightarrow \underline{\underline{86_{10}}}
$$

d) $0111\ 1001\ 1000_{BCD} + 0001\ 0100\ 0100_{BCD}$

$= 798 + 144 = 942_{10}$

Right nib

$$
\begin{array}{rr}
 & 1000 \\
+ & 0100 \\
\hline
 & 1100 \\
+ & 0110 \\
\hline
 & \underline{1}\ 0010
\end{array}
$$

Mid nibble.

$$
\begin{array}{rr}
 & {}^{1} \\
 & 1001 \\
+ & 0100 \\
\hline
 & 1110 \\
+ & 0110 \\
\hline
 & 1\ 0100
\end{array}
$$

Left nibble.

$$
\begin{array}{rr}
 & {}^{1} \\
 & 0111 \\
+ & 0001 \\
\hline
 & 1001
\end{array}
$$

$1001\ 0100\ 0010_{BCD}$

$\rightarrow \underline{\underline{942_{10}}}$

Question 5 |

ID = 64240072

a) 42.06_{10}

42 % 8 = 5, rem as 2

52_8

$0.06 \times 8 = 0.48 \Rightarrow 0$

$0.48 \times 8 = 3.84 \Rightarrow 3$

$0.84 \times 8 = 6.72 \Rightarrow 6$

fractional part (approximately)

↳ 0.036

final → 52.036_8

b) Convert to binary

$5_8 \to 101_2$

$2_8 \to 0102_2$

$0_8 \to 000_2$

$3_8 \to 011_2$

$6_8 \to 110_2$

101010.000011110_2

Question 1)

b) $F(A,B,C) = \sum(2,3,4,5,6)$

c) $\bar{A}B\bar{C} + \bar{A}BC + A\bar{B}\bar{C} + A\bar{B}C + AB\bar{C}$

d) $F(A,B,C) = \prod(0,1,7)$

e) $(\bar{A}+\bar{B}+\bar{C})(\bar{A}+\bar{B}+C)(A+B+C)$